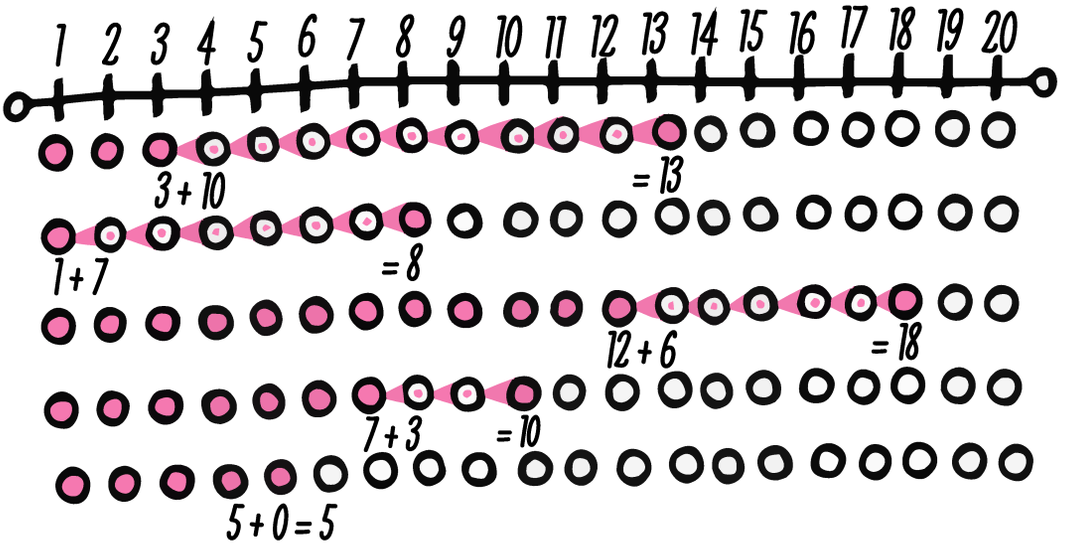
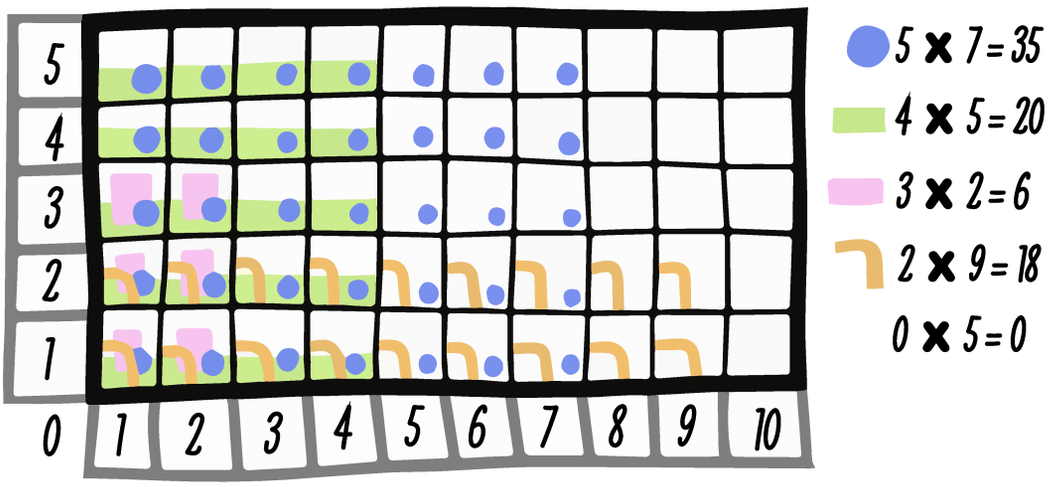
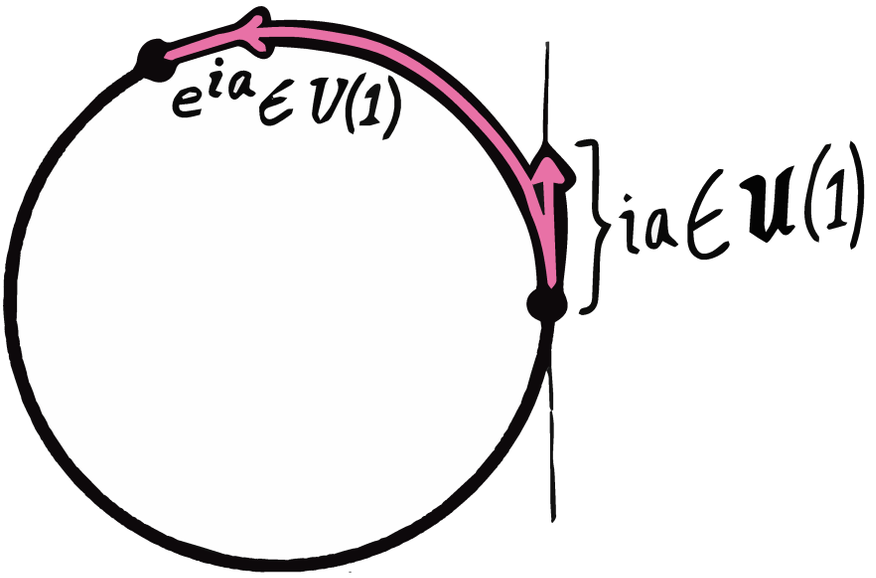
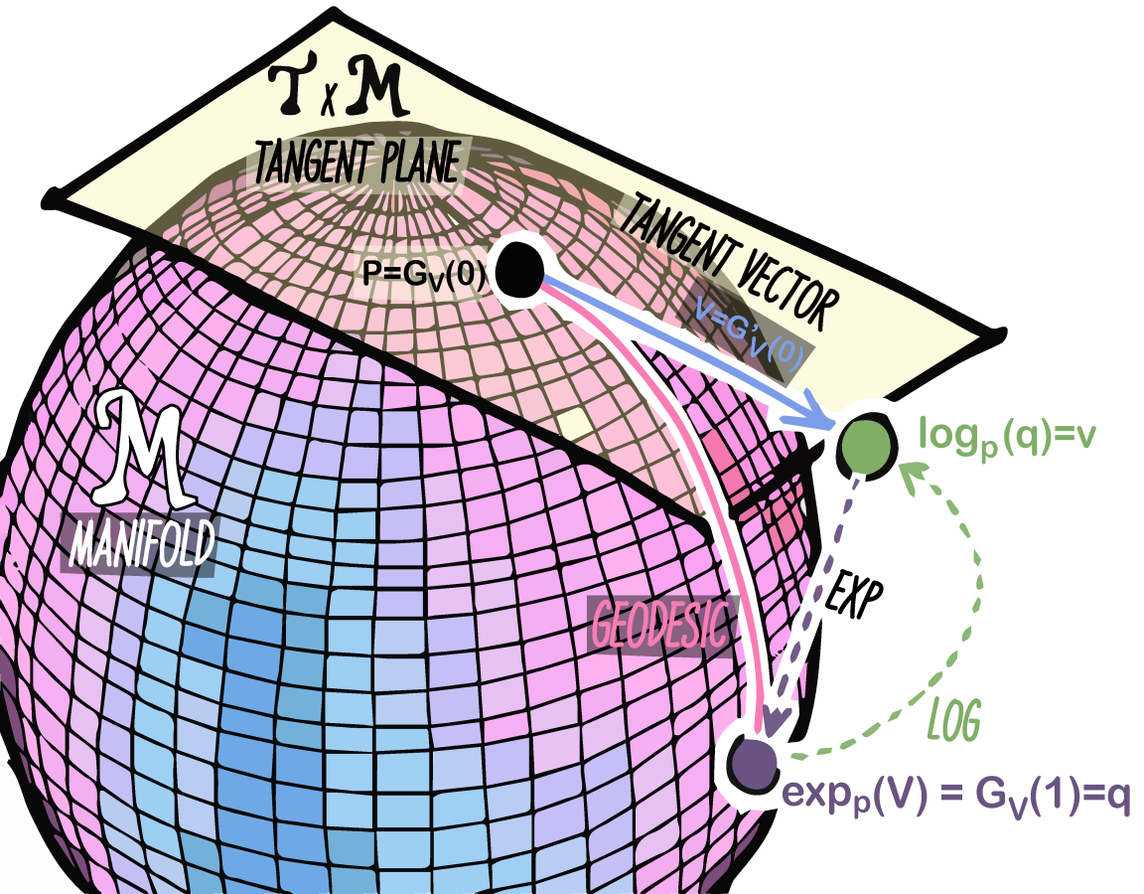

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
3+10
=13
1+7
=8
12+6
=18
7+3
=10
5+0=5


5
4
3
2
1
0
1 2 3 4 5 6 7 8 9 10
5 × 7 = 35
4 × 5 = 20
3 × 2 = 6
2 × 9 = 18
0 × 5 = 0


$e^{ia} \in U(1)$
$ia \in \mathfrak{u}(1)$


$T_x M$
TANGENT PLANE
TANGENT VECTOR
$P=G_V(0)$
$V=G'_V(0)$
$\log_p(q)=v$
M
MANIFOLD
GEODESIC
EXP
LOG
$\exp_p(V) = G_V(1)=q$